Technische Universität Braunschweig

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Strength of *wh*-island effects in subject vs. object relatives for L1/L2 English adults

Fred Zenker [1] and Bonnie D. Schwartz[2]

[1]Technische Universität Braunschweig

[2]University of Hawai‘i at Mānoa

fredzenker.github.io/gasla2026.pdf

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Gapped vs. resumptive relative clauses (RCs)

Gapped RCs (foot = unpronounced element)

the **class**$_1$ [that I'm thinking about dropping ___$_1$]

gap

head

foot

gapped RC dependency

- The primary relativization type in English

Resumptive RCs (foot = overt nominal)

*the **class**$_1$ [that I'm thinking about dropping **it** $_1$]

nominal

head

foot

resumptive RC dependency

- Widely held to be ungrammatical in all English RCs
- Used where gapped RCs are **difficult to process** or **inadmissible**

(Ferreira & Swets, 2005; Morgan & Wagers, 2018; Ross, 1967; Shlonsky, 1992)

Object vs. subject RCs

Object RCs (ORCs)

the **class**$_1$ [that the student dropped __$_1$]

- Foot in object position

Subject RCs (ORCs)

the **student**$_1$ [that __$_1$ dropped the class]

- Foot in subject position

Asymmetry in processing of English short-distance RCs: subject gaps easier than object gaps

(e.g., Just et al., 1996; King & Just, 1991; Lau & Tanaka, 2021)

Wh-island RCs

Syntactic islands: Domains filler-gap dependencies cannot enter

***Wh*-island:** Embedded clause with fronted *wh*-phrase

Wh-island ORC

*the **class**$_1$ [that I wonder [**which student**$_2$ __$_2$ dropped __$_1$]]

Wh-island SRC

*the **student**$_1$ [that I wonder [**which class**$_2$ __$_1$ dropped __$_2$]]

Asymmetry in processing of English island RCs: object gaps easier than subject gaps

(e.g., Han et al., 2012; Keffala & Goodall, 2011; Morgan & Wagers, 2018)

Morgan and Wagers (2018)

Whether-island conditions in the elicited production task

Type	Target response with **gapped** vs. **resumptive** RC
ORC	There's a prince$_1$ [that the ogre doesn't care [if the troll slayed {*__$_1$ /***him**$_1$}]].
SRC	There's a prince$_1$ [that the ogre doesn't care [if {*__$_1$ /***he**$_1$} slayed the troll]].

Written elicited production task

- More resumption in SRC condition than in ORC condition

Acceptability judgment task

- Lower ratings for gapped SRCs than for gapped ORCs

Conclusion: *Whether*-islands stronger in SRCs than in ORCs

The present investigation

- Part of a larger dissertation project (Zenker, 2024)
- Focus: subject–object asymmetry in *wh*-island RCs

  ! Not previously looked at with L2 learners (L2ers) of English

Participant groups

Adult L1-English controls (ENSs)

Adult L1-Korean L2ers of English (KLEs)

Adult L1-Mandarin L2ers of English (MLEs)

Studies

Participants distributed across two sub-studies: an ORC study and an SRC study

Why KLEs and MLEs?

Korean and Mandarin lack overt *wh*-movement in *wh*-questions

- Performance not traceable to L1 influence
- No help from target language input, either

English, Korean, and Mandarin compared

Grammatical gapped (–) and resumptive (+) RCs in environments tested

	Short-distance		Long-distance		*Wh*-island	
Language	SRC	ORC	SRC	ORC	SRC	ORC
English	–	–	–	–	Neither	Neither
Korean	–	–	–	–	N/A	N/A
Mandarin	–	±	±	±	N/A	N/A

Note. Based on available literature and judgments of our language consultants.

(Han, 2013; Keenan & Comrie, 1979; Morgan & Wagers, 2018; Pan, 2016)

English, Korean, and Mandarin compared

Grammatical gapped (–) and resumptive (+) RCs in environments tested

	Short-distance		Long-distance		*Wh*-island	
Language	SRC	ORC	SRC	ORC	SRC	ORC
English	–	–	–	–	Neither	Neither
Korean	–	–	–	–	N/A	N/A
Mandarin	–	±	±	±	N/A	N/A

Note. Based on available literature and judgments of our language consultants.

(Han, 2013; Keenan & Comrie, 1979; Morgan & Wagers, 2018; Pan, 2016)

English & Korean: Only gapped RCs in the short-distance and long-distance environments tested

English, Korean, and Mandarin compared

Grammatical gapped (–) and resumptive (+) RCs in environments tested

	Short-distance		Long-distance		*Wh*-island	
Language	SRC	ORC	SRC	ORC	SRC	ORC
English	–	–	–	–	Neither	Neither
Korean	–	–	–	–	N/A	N/A
Mandarin	–	±	±	±	N/A	N/A

Note. Based on available literature and judgments of our language consultants.

(Han, 2013; Keenan & Comrie, 1979; Morgan & Wagers, 2018; Pan, 2016)

Mandarin: Optional resumption in 3/4 of the same environments

English, Korean, and Mandarin compared

Grammatical gapped (–) and resumptive (+) RCs in environments tested

	Short-distance		Long-distance		*Wh*-island	
Language	SRC	ORC	SRC	ORC	SRC	ORC
English	–	–	–	–	Neither	Neither
Korean	–	–	–	–	N/A	N/A
Mandarin	–	±	±	±	N/A	N/A

Note. Based on available literature and judgments of our language consultants.

(Han, 2013; Keenan & Comrie, 1979; Morgan & Wagers, 2018; Pan 2016)

Wh-islands: English allows neither gapped RCs not resumptive RCs; that environment does not exist in Korean and Mandarin

Research questions (RQs)

RQ1: Does the subject–object asymmetry in the strength of *wh*-islands hold for adult L2 learners of English whose L1 lacks *wh*-movement in *wh*-questions?

RQ2: Is the strength of the asymmetry modulated by L2 English proficiency?

Participant background information

Participants in ORC and SRC studies prior to exclusions

Study	Group	*n*	Age at testing	C-test score	Age of onset
ORC	ENS	90	29.98 (18–71)	42.74 (26–49)	—
ORC	KLE	69	26.29 (18–41)	29.29 (7–45)	9.35 (8–15)
ORC	MLE	76	28.14 (18–45)	28.85 (7–49)	9.71 (8–14)
SRC	ENS	61	26.43 (18–58)	42.85 (22–50)	—
SRC	KLE	66	25.38 (19–36)	32.05 (11–46)	9.23 (8–13)
SRC	MLE	73	29.51 (18–46)	28.80 (11–48)	10.29 (8–14)

Note. Values are means and ranges; the C-test (max = 50) tested English ability.

Sequencing of tasks in the ORC and SRC studies

1. Language background questionnaire
2. **Oral elicited production task**
3. Self-paced reading task
4. **English acceptability judgment task**
5. Korean/Mandarin acceptability judgment task (L2 groups only)
6. English proficiency C-test (Zenker, 2024)
7. Exit survey

Oral elicited production tasks (EPTs)

- Conditions short-distance, long-distance, and ***wh*-island RCs**
- 15 critical items (3 conditions × 5 tokens, Latin-squared)
- 15 fillers eliciting short-distance SRCs with biclausal sentences

***Wh*-island conditions in the EPTs**

Study	Target response with **gapped** vs. **resumptive** RC
ORC	the man$_1$ [that Mary wonders [which officers$_2$ __$_2$ arrested {*__$_1$/*__**him**__$_1$__} last week]]
SRC	the man$_1$ [that Mary wonders [which book$_2$ {*__$_1$/*__**he**__$_1$__} borrowed __$_2$ last month]]

EPT example trial in the SRC *wh*-island condition

Mary wonders which book this man borrowed last month. ← stimulus sentence

Lisa wonders which book this man borrowed last month. ← foil sentence

Which man is this? ← question

Lisa wonders which book this man borrowed last month.

Recording...

See Ferreira and Swets (2005) for a similar procedure

"the man that Mary wonders which book {__/**he**} borrowed last month"

EPT response categories

1. **Gapped RC**
2. **Resumptive RC**
3. **Other RC**
4. **Nontarget**

EPT data exclusions

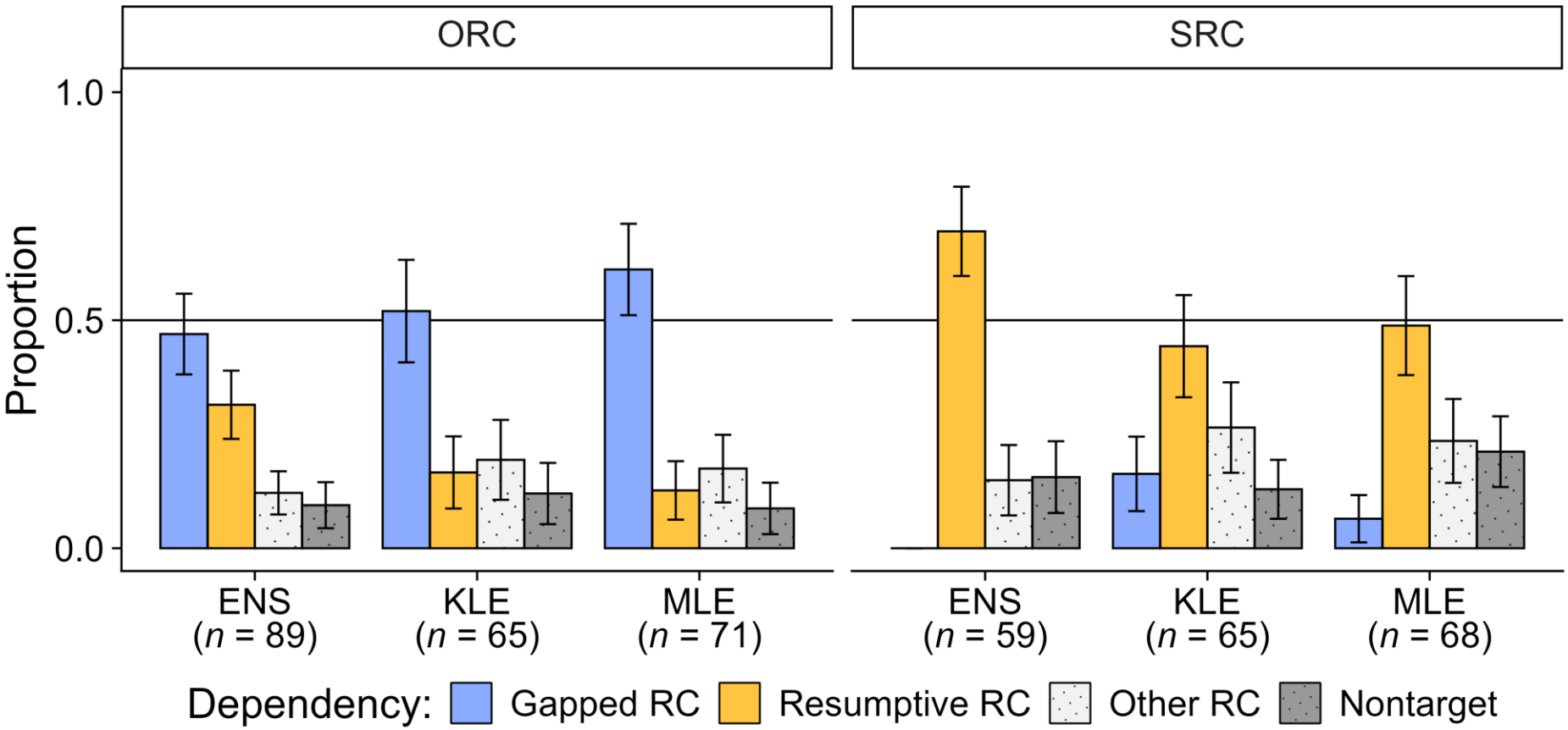
Experimental error

- ORC study: N/A
- SRC study: 1 MLE

Participants with only nontarget responses

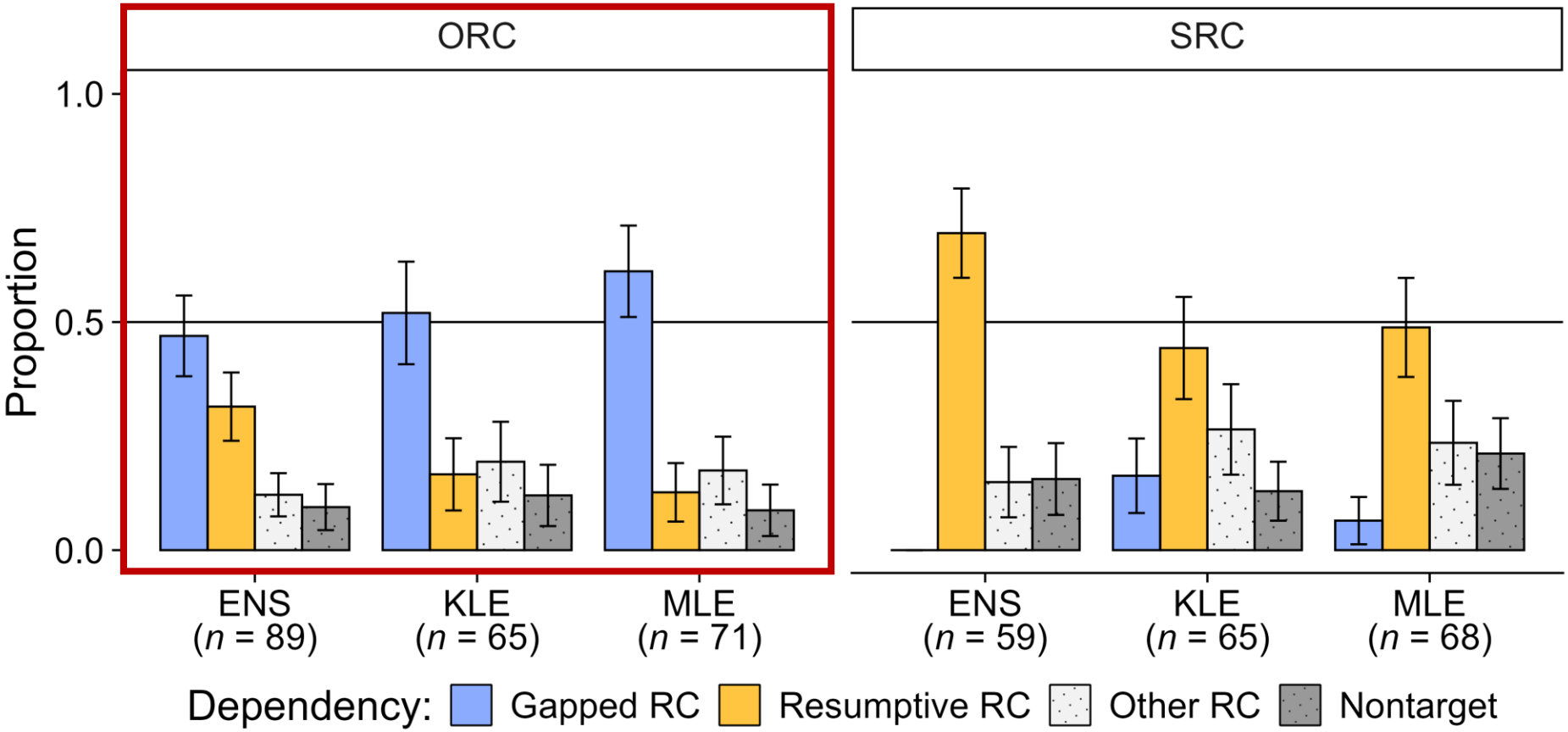
- ORC study: 1 ENS, 4 KLEs, 5 MLEs
- SRC study: 2 ENSs, 1 KLE, 4 MLEs

EPT results: Response rates by study and group



Note. Error bars are 95% confidence intervals.

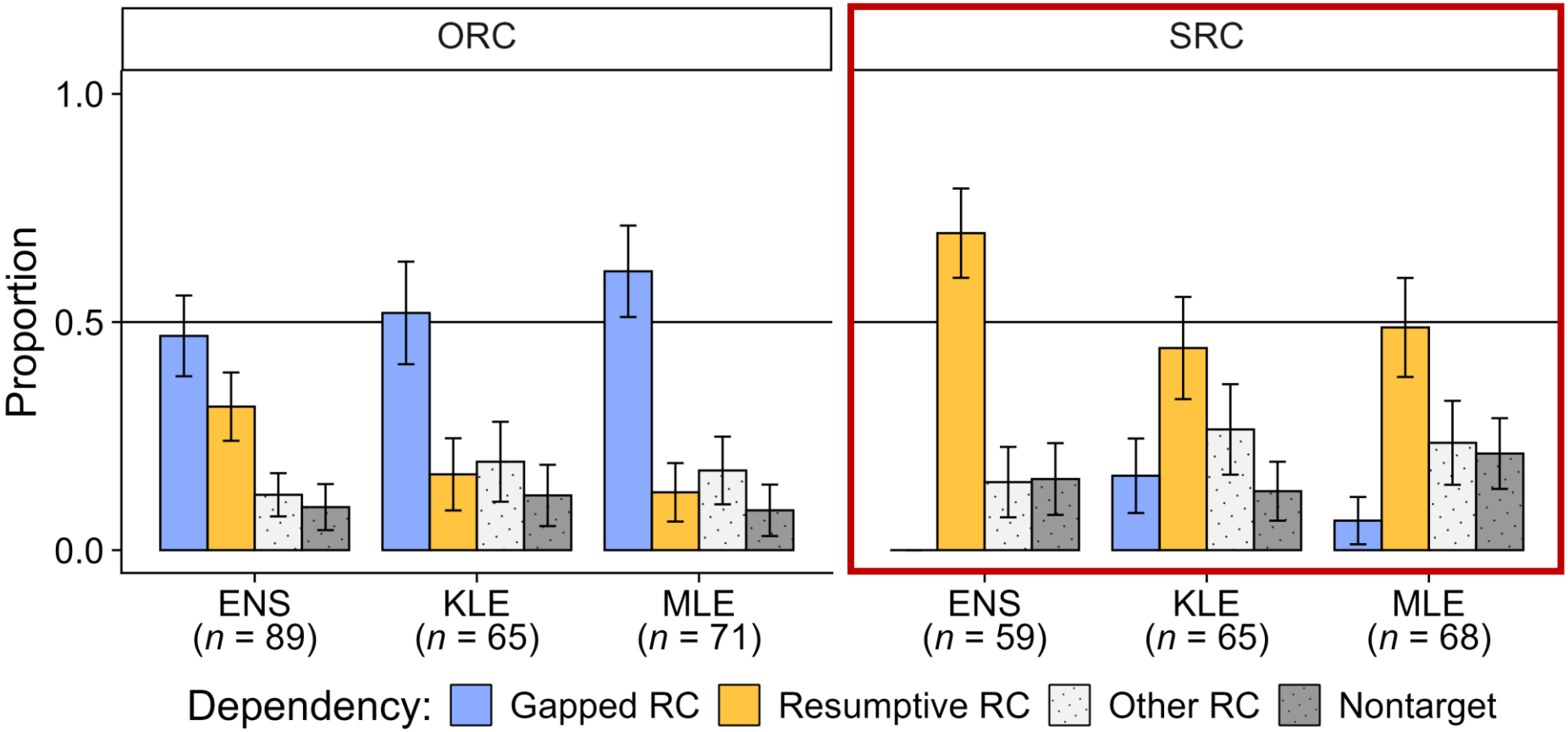
EPT results: Response rates by study and group



Note. Error bars are 95% confidence intervals.

ORC study: Gapped RCs most common

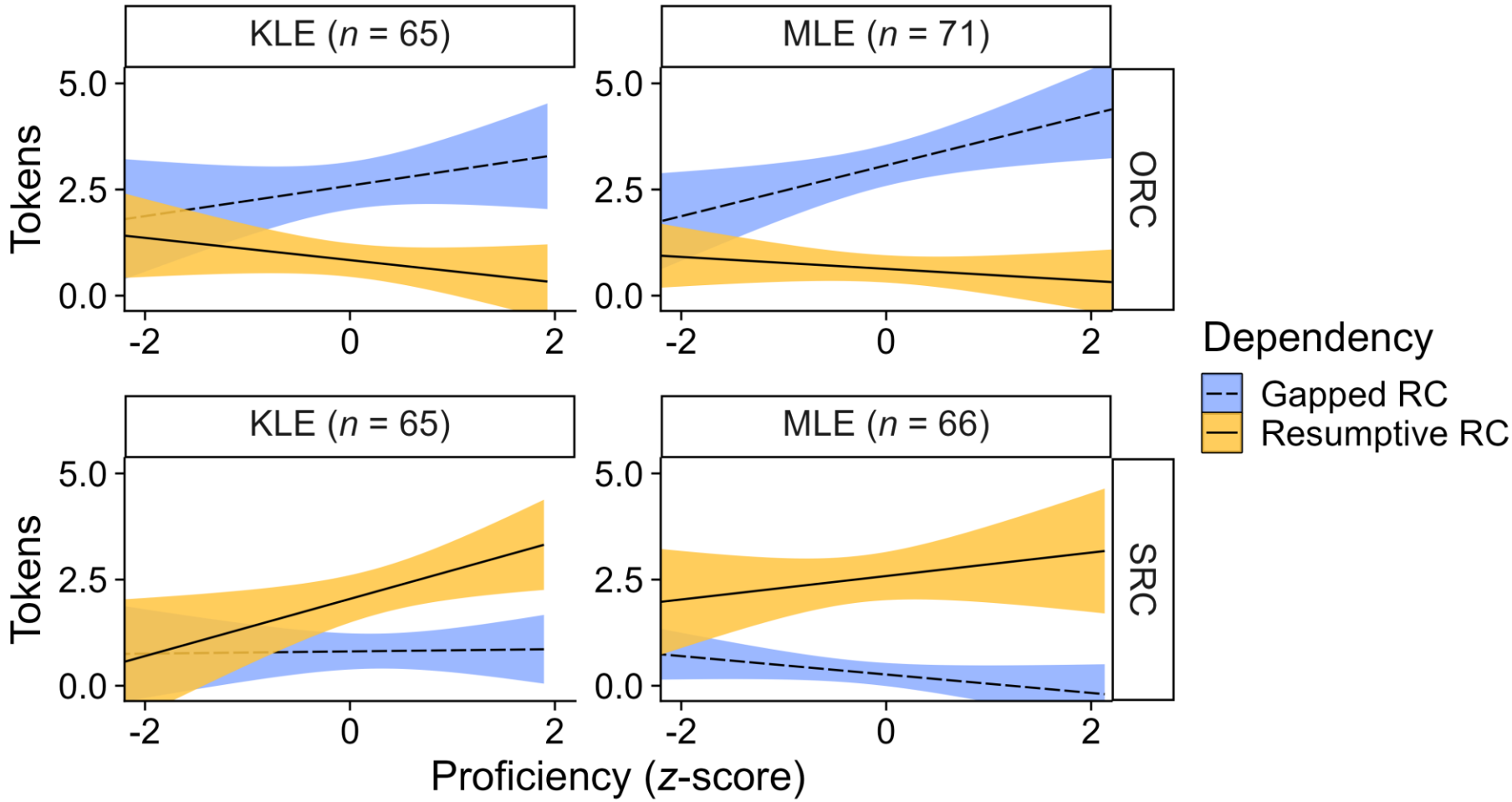
EPT results: Response rates by study and group



Note. Error bars are 95% confidence intervals.

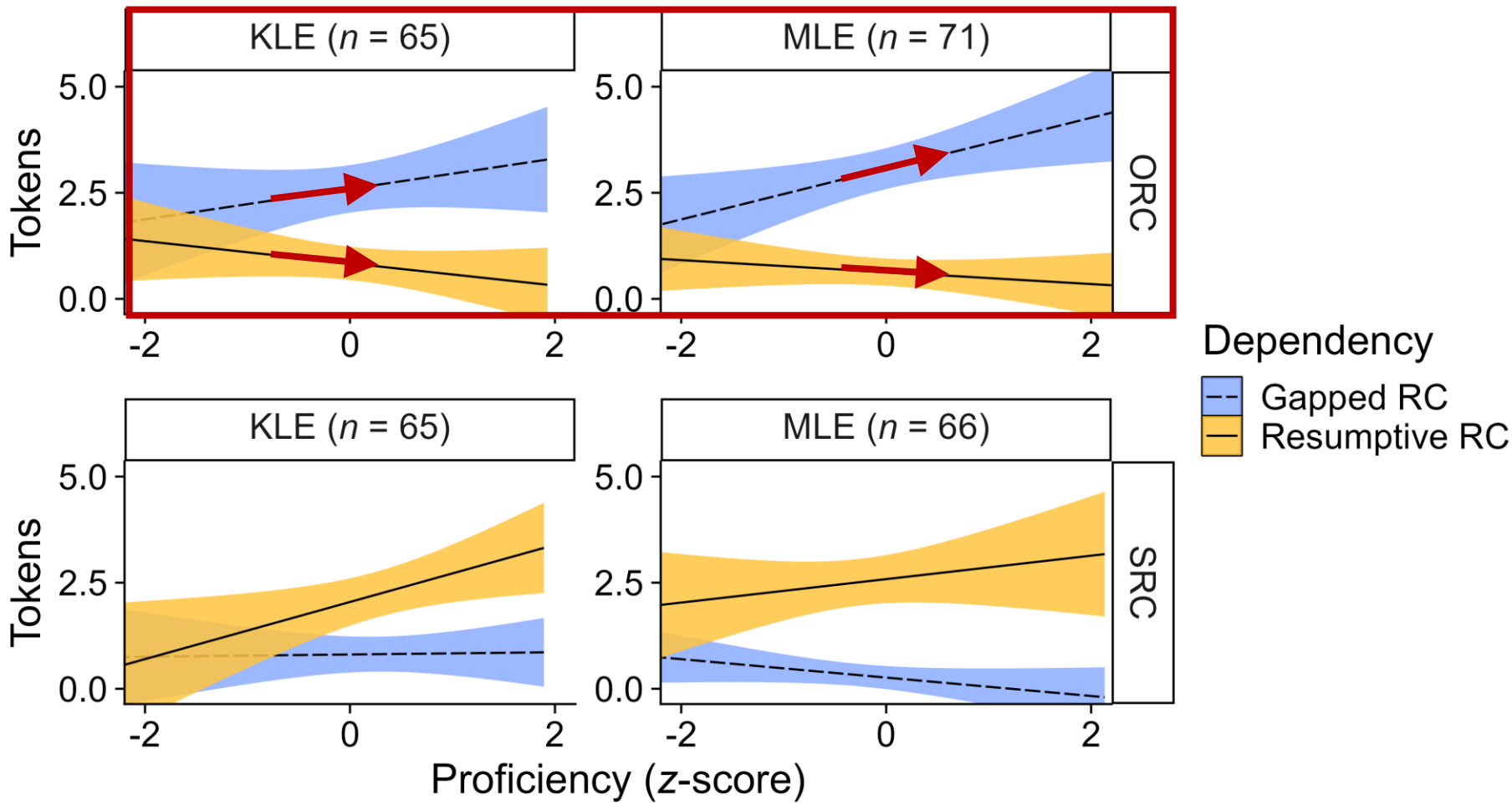
SRC study: Resumptive RCs most common; gapped RCs least common

EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals.

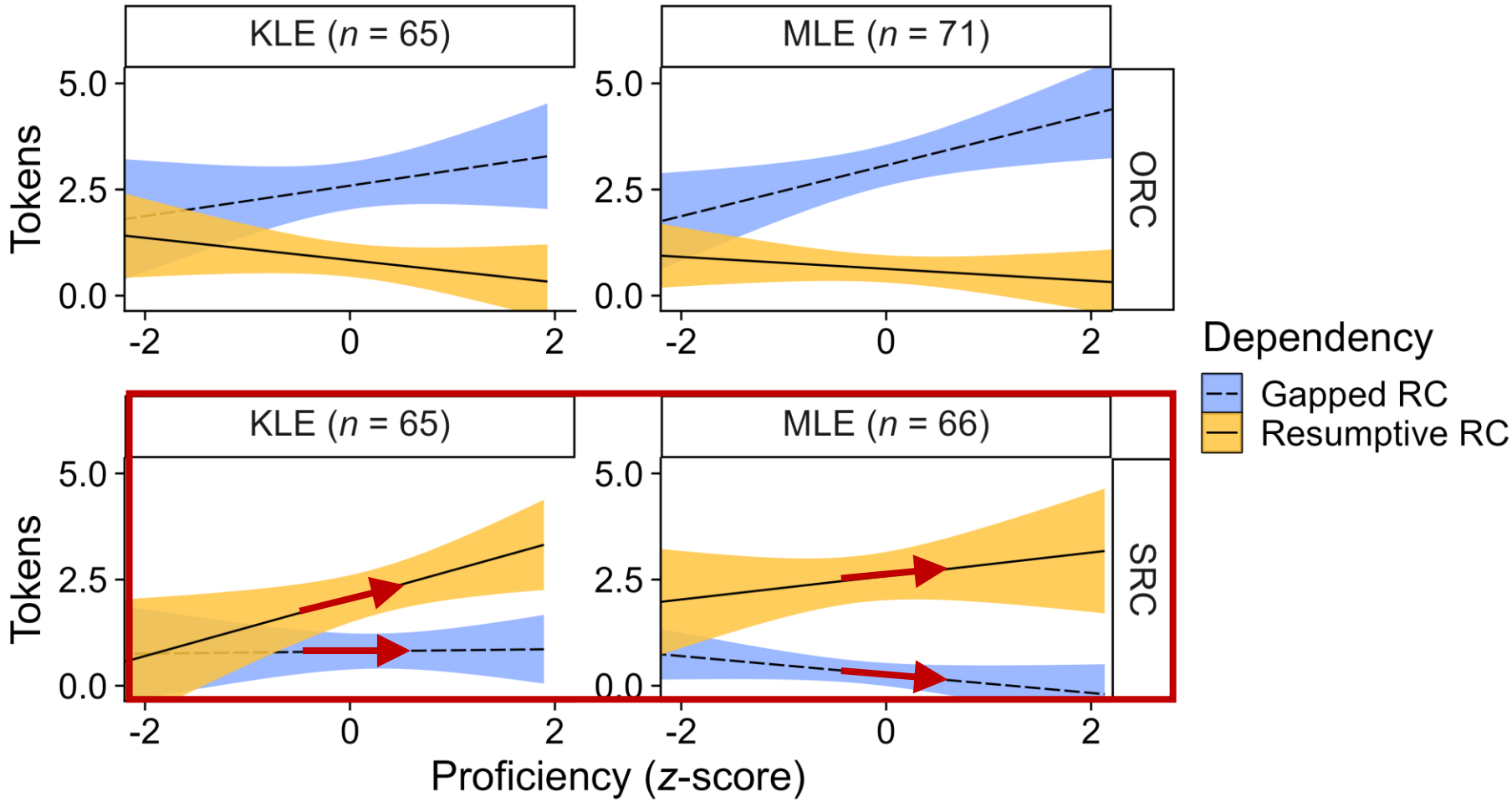
EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals.

ORC study: Higher proficiency associated with an emerging preference for gapped RCs over resumptive RCs

EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals.

SRC study: Higher proficiency associated with an emerging preference for resumptive RCs over gapped RCs

English acceptability judgment tasks (AJTs)

- 2 × 3 design crossing DEPENDENCY (gapped RC; resumptive RC) and ENVIRONMENT (short-distance; long-distance; ***wh*-island**)
- 30 critical items (6 conditions × 5 tokens, Latin-squared)
- 42 fillers (28 grammatical; 14 ungrammatical)

Wh-island conditions in the AJTs

Study	Example stimulus with **gapped** vs. **resumptive** RC
ORC	Mary knows the man$_1$ [that I wonder [which officers$_2$ __$_2$ arrested {*__$_1$/*he**him**$_1$} last week]].
SRC	Mary knows the man$_1$ [that I wonder [which book$_2$ {*__$_1$/*he**he**$_1$} borrowed __$_2$ last month]].

AJT fillers

- 6 types (7 tokens each)
- Most and least acceptable types plotted alongside critical conditions for reference

Most and least acceptable filler types in the AJTs

Type	Example
"Grammatical"	Mary says [this man saw the criminals$_1$ [that __$_1$ attacked me last week]].
"Ungrammatical"	*This woman says [John met the protesters$_1$ [that __$_1$ were criticized me last week]].

Passive verb with extra argument

AJT example trial in the SRC *wh*-island condition

stimulus

Mary knows the man that I wonder which book he borrowed last month.

Completely Unacceptable 1 2 3 4 5 6 *Completely Acceptable*

Press the zero key if you cannot rate the sentence

6-point rating scale

I-don't-know option

AJT data exclusions

Experimental error

- ORC study: 1 KLE
- SRC study: 1 MLE

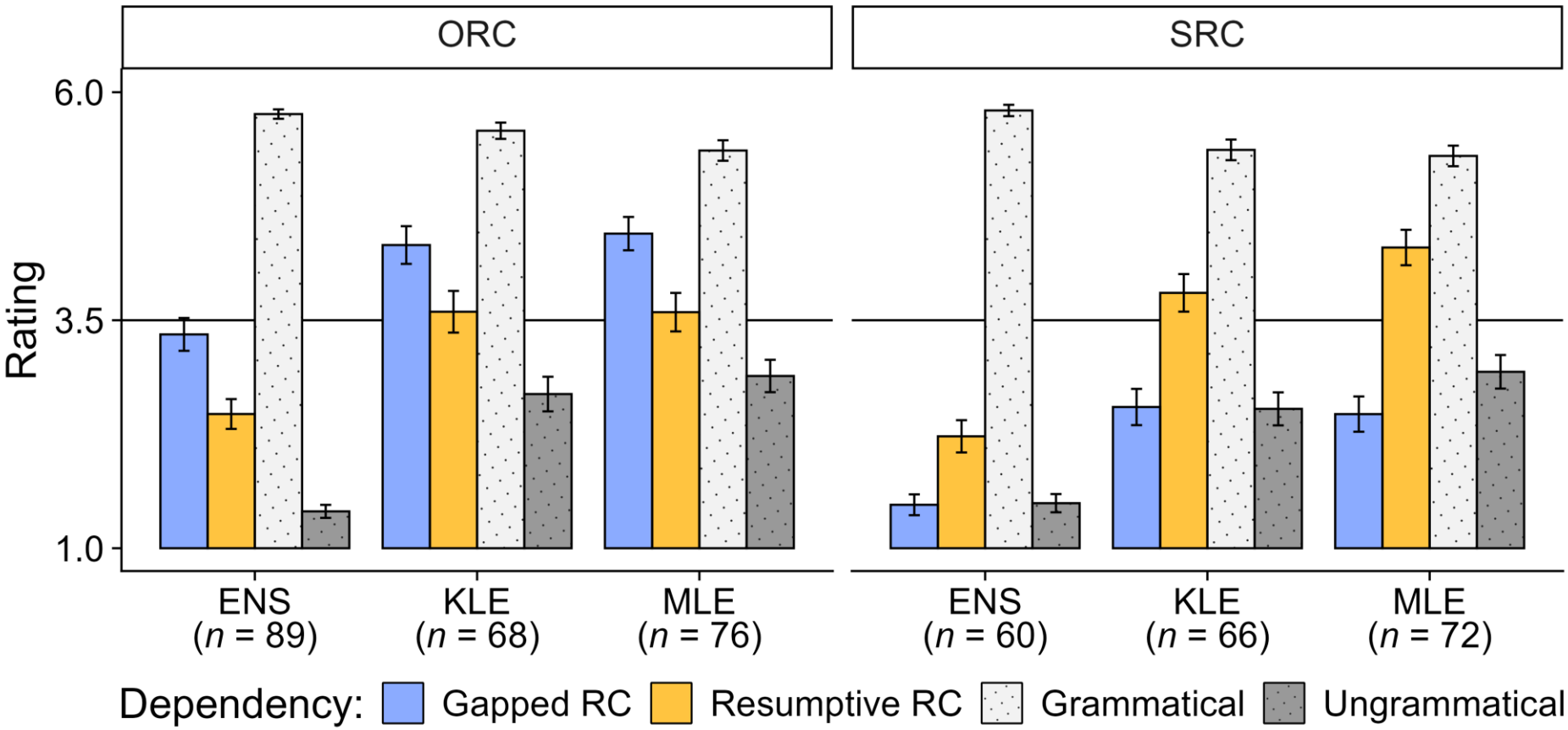
I-don't-know responses

- ORC study: 22 responses (0.13%), 7 from *wh*-island condition
- SRC study: 30 responses (0.21%), 10 from *wh*-island condition

***Yes*-bias outliers**

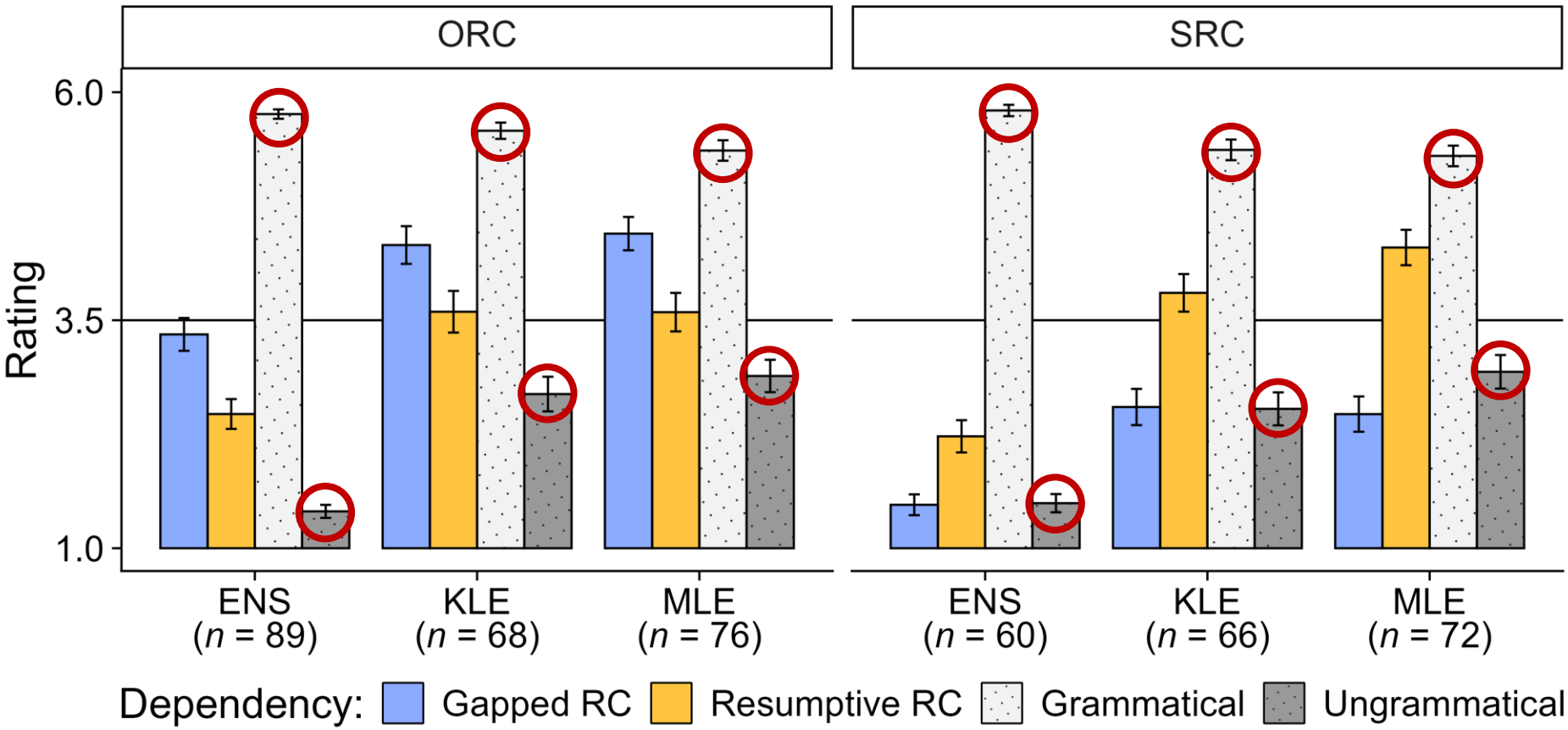
- ORC study: 1 ENS
- SRC study: 1 ENS

AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals.

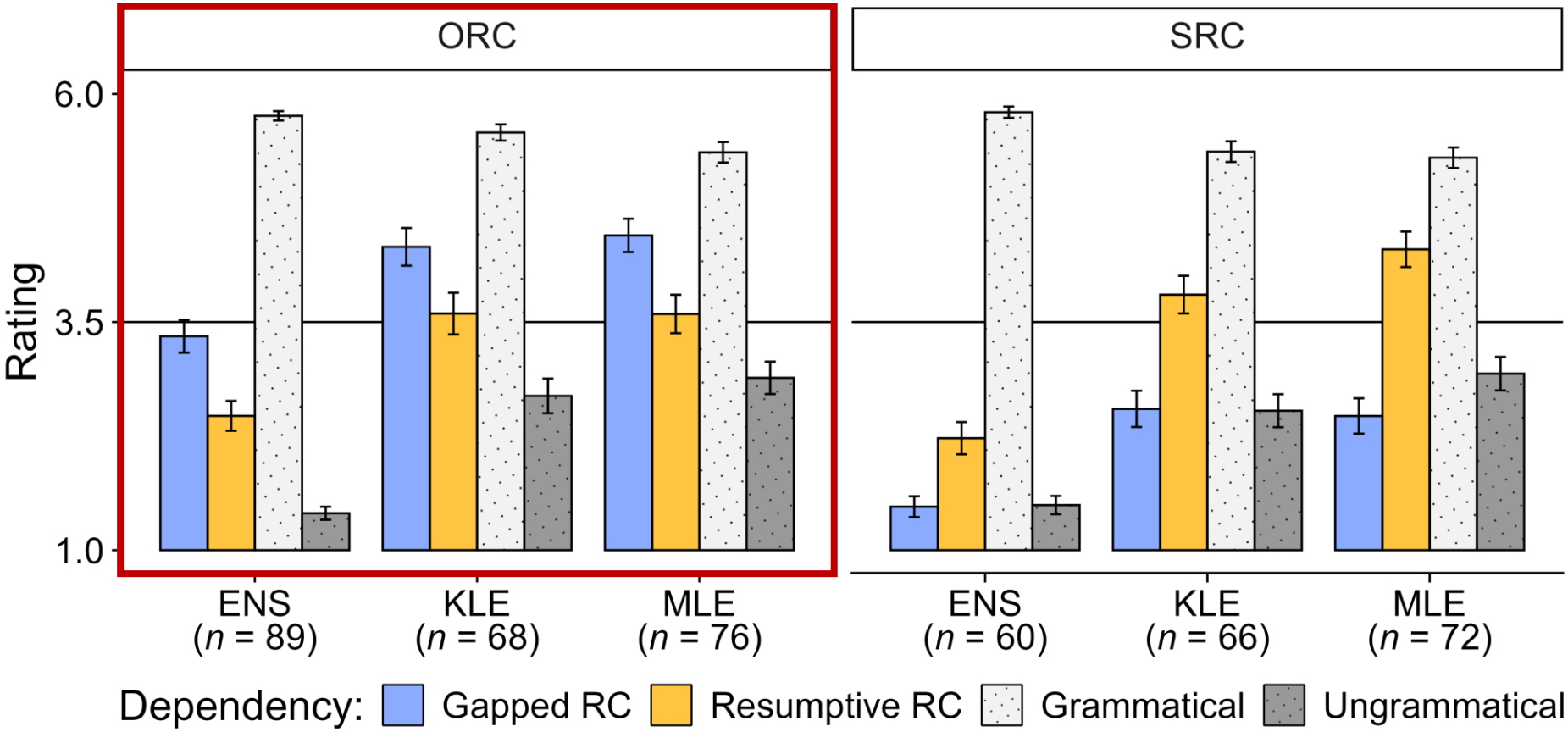
AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals.

Ratings for fillers patterned very similarly across the two studies

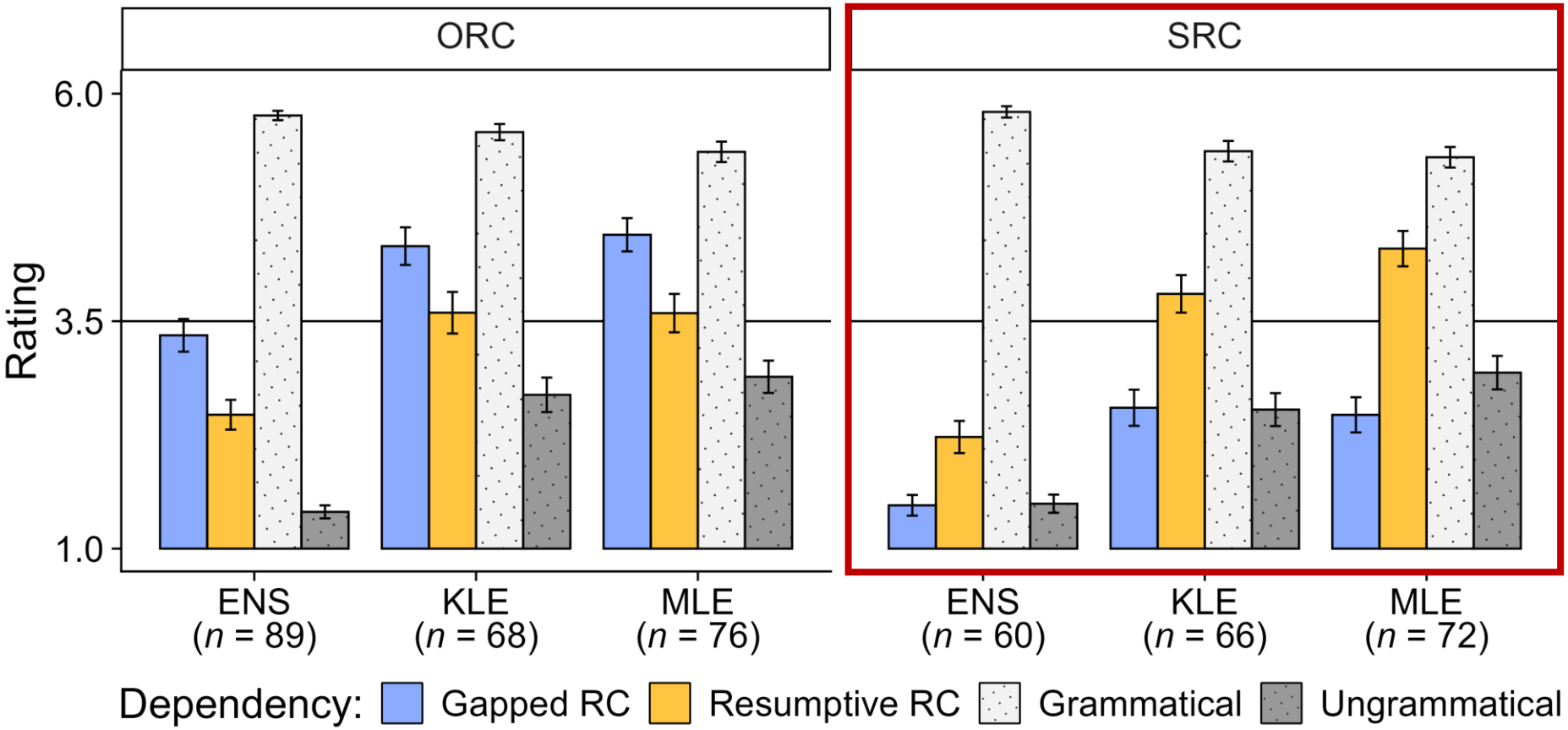
AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals.

ORC study: Gapped RCs rated higher than resumptive RCs; both are intermediate between grammatical and ungrammatical fillers

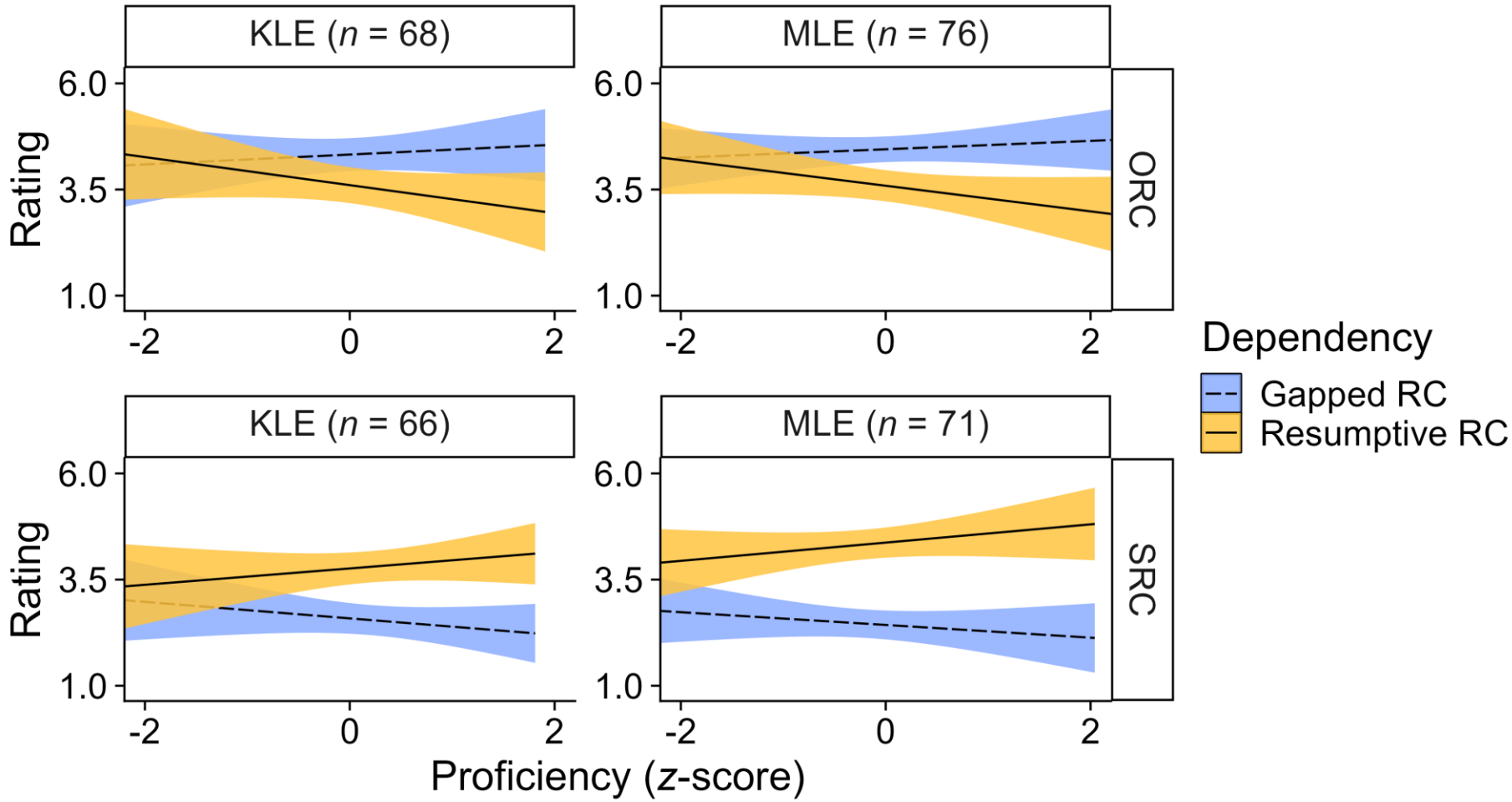
AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals.

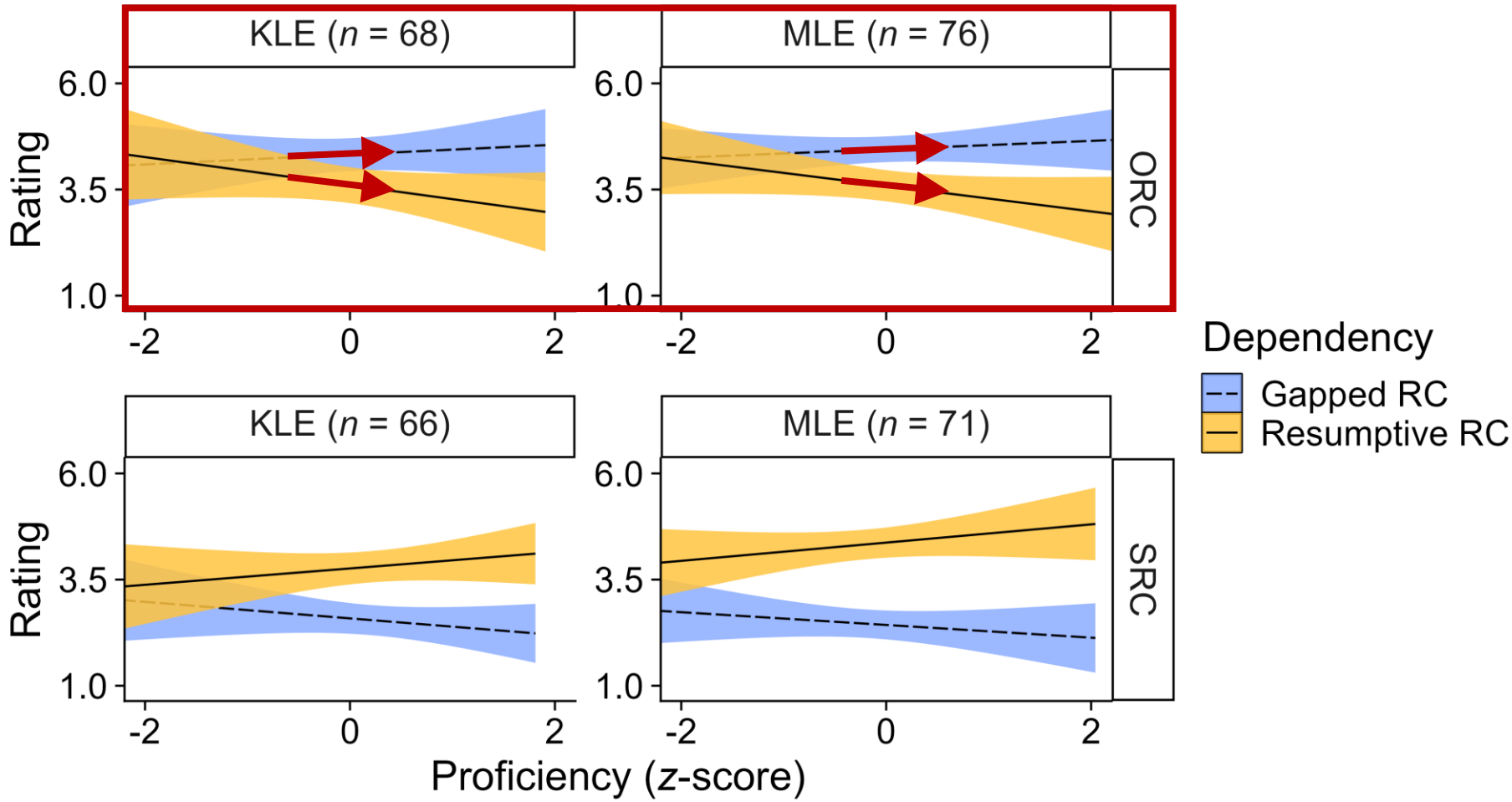
SRC study: Gapped RCs rated lower than resumptive RCs; on par with or lower than ungrammatical fillers

EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals.

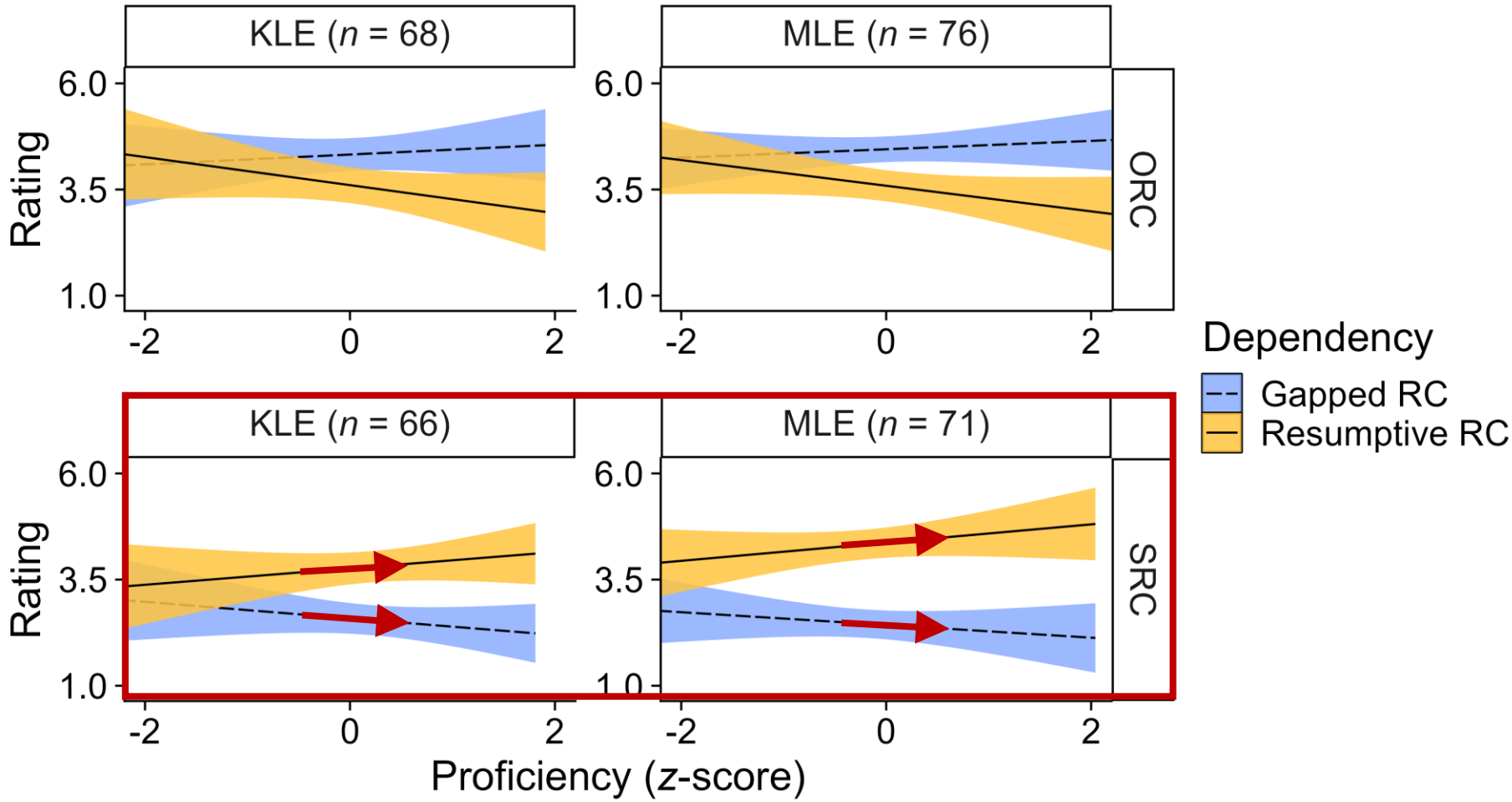
EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals.

ORC study: Higher proficiency associated with an emerging preference for gapped RCs over resumptive RCs

EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals.

SRC study: Higher proficiency associated with an emerging preference for resumptive RCs over gapped RCs

Summary of results

***Wh*-island ORCs**

- **All groups:** Gapped RCs produced more often and rated higher than resumptive RCs
- **L2 groups:** Rising proficiency associated with emerging preference for gapped RCs over resumptive RCs

***Wh*-island SRCs**

- **All groups:** Gapped RCs produced less often and rated lower than resumptive RCs
- **L2 groups:** Rising proficiency associated with emerging preference for resumptive RCs over gapped RCs

Research questions revisited

RQ1: Does the subject–object asymmetry in the strength of *wh*-islands hold for adult L2 learners of English whose L1 lacks *wh*-movement in *wh*-questions?

Yes, results from all groups indicated *wh*-island effects are stronger in SRCs than in ORCs

RQ2: Is the strength of the asymmetry modulated by L2 English proficiency?

Yes, KLEs and MLEs both showed proficiency effects whereby increasing proficiency brought them more in line with native speakers

Accounting for the asymmetry in island strength

Two major types of explanations:

- Representational accounts
- Processing accounts

Representational account: ECP violation (Chomsky, 1981)

Subject-gap islands violate an extra grammar constraint, such as the Empty Category Principle (ECP)

ECP: Traces must be properly governed (Chomsky, 1981)

***Wh*-island ORC → Only violates island constraint (better)**

*the **person**$_1$ [**OP**$_1$ that I wonder [**which lawyer**$_2$ ***t***$_2$ hired ***t***$_1$]

✓ verb *hired* c-commands the object trace t_1

✓ no barriers intervene between *hired* and t_1

Proper government achieved

***Wh*-island SRC → Violates island constraint and ECP (worse)**

*the **person**$_1$ [**OP**$_1$ that I wonder [**which lawyer**$_2$ ***t***$_1$ hired ***t***$_2$]

✓ antecedent OP$_1$ c-commands subject trace t_1

✗ *wh*-phrase *which lawyer* is barrier between OP$_1$ and t_1

No proper government

(McDaniel & Cowart, 1999; Keffala & Goodall, 2011)

Processing account: Clause planning (McDaniel et al., 2015)

- Significant cognitive effort is needed for both…
  - **(a)** clause planning
  - **(b)** filler integration
- Overlapping spikes in cognitive load → acute difficulty

_Wh_-island ORC → non-overlapping spikes (easier)

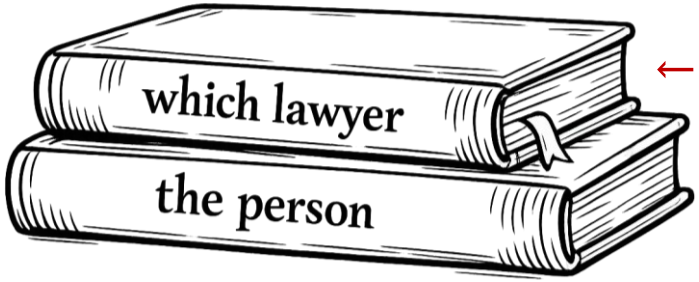
*the **man**$_1$ [that wonder [**who**$_2$ __$_2$ hired __$_1$ last week]
XXXXXXXX ← Planning
XXXXXXXX ← Integration

_Wh_-island SRC → overlapping spikes (harder)

*the **man**$_1$ [that wonder [**who**$_2$ __$_1$ hired __$_2$ last week]
XXXXXXXX ← Planning
XXXXXXXX ← Integration

Processing account: Push-down storage (O'Grady, 2005)

Push-down storage principle: Recently stored items accessed more readily than previously stored ones (Marcus, 1978)



← top of memory stack

← bottom of memory stack

***Wh*-island ORC → top item retrieved first (easier)**

*the **person**$_1$ [that wonder [**which lawyer**$_2$ __$_2$ hired __$_1$]

***Wh*-island SRC → bottom item retrieved first (harder)**

*the **person**$_1$ [that wonder [**which lawyer**$_2$ __$_1$ hired __$_2$]

Conclusion

- *Wh*-island effects in English are stronger in SRCs than in ORCs
- L1-Korean and L1-Mandarin L2ers of English can become sensitive to the asymmetry even without help from the L1
- The difference in island strength comes into focus with rising L2 proficiency
- More research needed to determine whether representational accounts or processing accounts provide the best explanation

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Software

Experimental software:

The experiments were built using the jsPsych library (de Leeuw, 2015) and hosted on Cognition (https://www.cognition.run). Participants were recruited on Prolific (https://www.prolific.com), on Korean online job boards such as SNULife (https://www.snulife.com) and Spec Up (https://cafe.naver.com/specup), on Taiwan's PTT bulletin board system (https://www.ptt.cc), and through the Language Analysis and Experimentation Labs at the University of Hawaiʻi at Mānoa (https://sites.google.com/a/hawaii.edu/laelabsedit).

Data analysis tools:

All analyses were performed in R (R Core Team, 2024). Mixed-effect models were fit using the *lme4* (Bates et al., 2017) and *ordinal* (Christensen, 2019) packages. Post-hoc testing was done with the *emmeans* package (Lenth, 2022).

Image credits:

This presentation uses images generated in ChatGPT (https://chatgpt.com) and icons from Flaticon (https://www.flaticon.com).

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감사합니다 谢谢

Thank you!

Mahalo Danke

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